

AN OVERVIEW OF PREVIOUS STUDIES ON THE PERFORMANCE OF MACHINE FOUNDATION UNDER NUMEROUS DYNAMIC VARIABLES

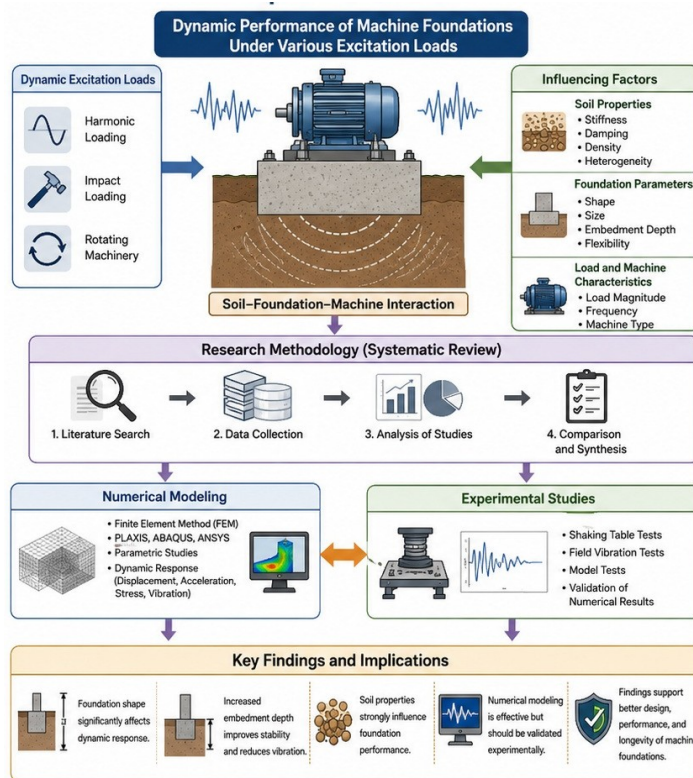
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Graphical abstract



Abstract

Dynamic performance of machine foundations under various excitation load is one of the important problems in engineering. Machine foundations dynamic response affected by several parameters including machine and soil properties, loading magnitudes and geometries, foundation geometry, shape, and embedded depth. A systematic methodology was adopted to collect and analyze significant studies. Through an extensive review of the literature, the discussed papers illustrate the contributions and effects of soil properties, improvement, techniques, foundation types, and load conditions on performance. Several studies have confirmed that the foundation shape and embedment depth are among parameters which extensively affect its dynamic behavior. Investigation regarding the effect of both dynamic loading and soil conditions is needed. While numerical Modeling simulations are considered as an effective tool for studying and determining foundation dynamic response, they should be supported by experimental studies to obtain a clear understanding. These outcomes offer useful inputs for the machine foundation design phase, and subsequently lead to better performance and longevity from a dynamic perspective. Addition to the identified research gaps with respect of different influencing factors, heterogeneous field conditions These findings help identify potential mechanisms and processes to achieve better design and performance of machine foundations exposed to dynamic loading.

Keywords: Machine foundations, Dynamic response, Soil-structure interaction, Vibration analysis, Embedment depth.

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1.0 INTRODUCTION

Machine are considered a key part in factories, power plants and others industry activities. The machine should be supported by adequate foundation to withstand the vibration forces and transform them to the soil without any risk. Experimental and theoretical techniques can be used to make a great deal of contribution to this knowledge. Therefore, general computational numerical modeling has

mostly been used in simulation the responses of foundations to dynamic loads. In using the finite element method, the material properties and geometries can be simulated accurately, it is suitable for analyzing the machine foundation behavior under exposure of loads. The numerical simulation methods provide worthy valued information on the behavior of soil, and the response of the foundation under the dynamic loads. Moreover, the experimental and numerical approach provides the

investigator with a better vision of the role of vibration in relation to the soil than the other approach. The acting loads on the foundation might be static or a combination of static and dynamic. Whereas there are various dynamic force sources, the determination of foundation frequency and amplitude resonant has drawn much attention in the recent years, especially in the design of machine foundations, and the seismic design of large structures like nuclear power plants. Investigating the dynamic behavior of the structural foundations under different vibration modes is the critical problem in the design of machine foundation and the dynamic soil structure interaction problem [1].

In this manner, when using experimental or numerical models, improvements of machine foundation design, construction as well as maintenance become possible to enhance its performance and reliability. Machine foundation design developed in the early twentieth century through main contributions by researchers such as Geiger and Rauch, who concentrate, on natural frequencies and vibration problems. Future, Timoshenko, Den Hartog, and D.D. Barkan developed vibration analysis and the study of dynamic effects, with Barkan's 1962 book investigated dynamic effects on machine foundations becoming a fundamental reference in the field.[2]. An overall review of the dynamic behavior of foundations supporting vibrating machines, presenting theoretical concepts such as frequency, resonance, damping ratios, and soil-structure interaction, was provided. The work confirms the necessity of numerical models in expecting how different vibration modes affect foundation stability [3].

Dynamic performance design of wind turbine machine foundations based on specific site characteristics subjected to vibrations analysis involving three cases using finite element analysis. The research demonstrated the significance of confirming analytical solutions with finite element simulations. However, an often miss aspect of machine foundation design is the post construction evaluation of foundation performance and its comparison with the expected behavior. Evaluation of post construction and continuous observation is necessary to improve the accuracy of analytical solutions and finite element analysis during the design process [4].

More recently, the application of software to realize theoretical principles (Barkan method) was employed for designs of block foundations type. In this respect, it is a hybrid approach between traditional theory and modern computational tools that allows engineers to simulate and optimize foundation performance prior to construction via numerical simulation. Machine-induced vibrations forms the basis for understanding of the complex interaction between soil and foundation during dynamic process or excitation [5].

In order to produce electrical energy, machine generators are often used at home which they also generate dynamic vibrations during their operation. When these generators are placed, without proper site assessment, they can have an adverse impact on near foundations and adjoining soil. These vibrations lead to the foundation settlement and crack or fracture risk in the structures as well as the machine itself. Moreover, if the foundations are not planned to withstand these dynamic

loads, then over a period of time, adjacent buildings may not be preserved. Assessment of the local soil characteristics is an essential factor since these characteristics govern the structure bearing capacity and settlement. To provide the properties that can perform under different conditions, safe and efficient design of this type of foundation is addressed with the help of geotechnical investigations. Due to the increasing amount of construction in areas subject to vibrations of various types, the problem of controlling these vibrations becomes increasingly relevant. Their continuous action on structures causes material loss [6].

Dynamic loads may arise from different sources, which vary over time and they can generate vibrations in the foundation. Vibration may have an impact on foundation causing resonance which happens when the natural frequency of the foundation relates to the frequency of the machine. Excessive displacement caused, structural damage, and ultimately the failure of the foundation. By utilizing isolation units such as vibration isolators or damping materials that can absorb and dissipate vibratory energy, greatly it possible to minimize the effects of dynamic loads by making sure that the natural frequency of the foundation is not interference with the operating frequencies of machine. The characteristics of the loads generated by machinery operation significantly affect the behavior of foundations. Vibrations generated by rotating machinery can cause numerous problems for adjacent structures. [7].

The magnitude of the excitation loads should be taken into account during analysis and design of the machine foundation. Static loads are relatively simple compared to dynamic loads, which the later vary with time and direction. Proper assessment of these load characteristics ensures that the foundation can sustain operational stresses without excessive deformation or failure. Dynamic forces can be either self-induced forces due to the operation of the machine itself and/or directly applied [8] The components of a common machine foundation comprises from (i) machine: such as rotary, reciprocating, impact machines (ii) foundation: such as block or frame (iii) support medium: soil layers. Due to progression of technology, high-speed machines have been established. With the increase in machines speed, the effect of vibrations grow, negatively impacting the machine foundation system. Figure (1) clarify a brief overview of the machine foundation types [9][10].

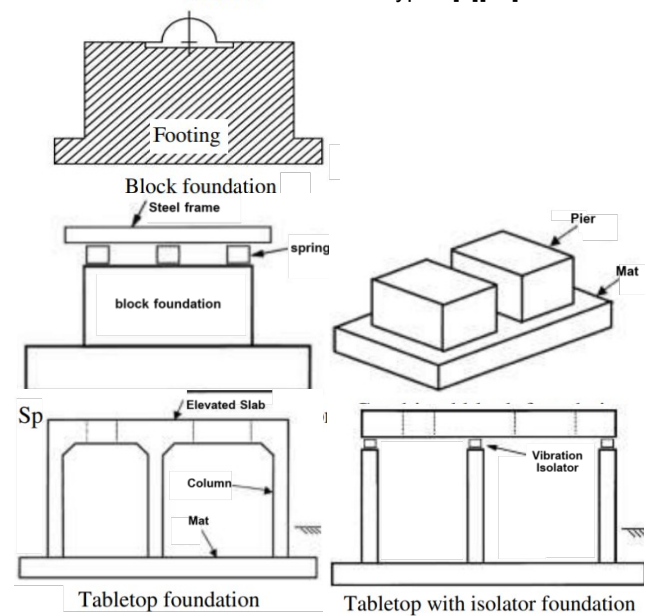


Figure (1): Machine foundation types [9][10]

Foundations are crucial in supporting machines and minimizing their dynamic responses to vibrations. **Block foundation** is suitable for heavy machines with low natural frequency. The effectiveness depends on soil properties, with the dynamic response being influenced by load, mass, dimensions, and surrounding soil properties. **Box-type foundations**, used for machines with lower mass and higher natural frequency. Due to its hollow structure, this foundation has less mass than block-type foundation, resulting in a higher natural frequency [9][10].

For situations where multiple machines are located close together, **combined foundation** used. Such foundation is similar to simple block foundation but is designed to support closely spaced machines. The design is more challenging due to the combined forces created from multiple machines and the lack of stiffness in a larger foundation mat [9][10].

Spring-mounted block foundation is used for high-vibration machines and to isolate thermal forces. It consists of vertical columns and a horizontal frame, with the machine supported within the frame. This design is highly effective in isolating vibrations and thermal forces. **Tabletop foundation** is usually used for turbine foundations, such as electric generators. It allows ducts, piping. **Tabletop with isolator foundation**, may have a complex response to dynamic loads. However, this type of foundation used for high-vibration machines like pumps, blowers, and fans. The supporting columns help to reduce the response of dynamic loading, and their effectiveness depends on the machine's frequency and speed [9][10]

2.0 METHODOLOGY

This study adopts a systematic review approach to explore the performance of machine foundations under dynamic variables. Relevant studies were collected from major scientific databases, including Scopus and Web of Science. An extensive literature search was performed with a combination of keywords like machine foundation, dynamic response, vibration analysis, soil-structure interaction and foundation performance. Inclusion criteria included: (i) peer-researched journal articles, (ii) investigations involving machines subjected to dynamic loading, (iii) related studies with respect to soil condition, and foundation type, or vibration type. The exclusion criteria included: Exclusion criteria included: (i) conference papers, (ii) theses and dissertations, (iii) non-English publications, and (iv) studies not directly related to machine foundation performance. Duplicates and irrelevant studies were removed after the initial search. Out of these, the articles were screened as per title, abstract and full-text review to check if they lie underneath the scope of this study.

3.0 Overview Machine Foundations and their Significance in Engineering

Machine foundations are very important in many fields specifically in determining the sufficiency and accurate functioning of the machine where the vibrations are expected. The study of these foundations design is essential to mitigate the influence of vibrations and operating forces. As the interaction between the machine and foundation influences overall system performance, engineers must consider various factors, including soil properties, load types, dynamic responses, and machine characteristics. Machine foundation interaction system is defined as a combined system consisting of machine, supporting foundation, and soil layer on which the foundation rests, as illustrated in the Figure (2).

The design of machine foundation requires special consideration because they transfer dynamic loads to the soil in addition of the loads due to the weight of the machine foundation system. The dynamic load due to the operation of the machine is generally small compared to the static weight of the machine plus the foundation. When the machine is operating, the cyclic load is generated over a long period with a small magnitude; so, if the soil behavior is not elastic, deformation will increase with each loading cycle [12].

Figure (2): Machine foundation system [11]

The reliable design of the machine foundation allowing for engineers to minimize the effect of vibrations induced by the machine. Several studies were proving the design and an estimation of the capability of the machine foundation unit's working with every reciprocating or rotating machine conversely, various parameters and soil properties can contribute these vibrations and significantly affect adjacent structures [13][14].

The transmission of machine vibration to the foundation and then to the soil is one of the most important engineering issues presented in the substructure of machines which foundations are based on the soil [15][16] [17]. Machine foundations are one of the basic components for any structure found in a multitude of industrial structures, as machinery foundations, fertilizer plants, power plants, petrochemical complexes, and steel plants. The vibration of a machine is related to the great settlement of its foundation which can influences the

function of the machines. The assessment of the machine which was supported by the structural members required an understanding of the mechanical characteristics of the machine as well as the structural characteristics of the foundation and soil. The foundation parameters which affect the machine-foundation vibrations system mainly include: foundation size, embedment depth, applied dynamic forces, dynamic parameters of soil. Foundations that support machines require dynamic analysis where the calculation of the natural frequency of the system and some essential soil parameters such as damping and stiffness are essential. Dynamic analysis complements static analysis which is necessary to ensure the foundation's strength against excessive settlement [15] [16][17].

The analysis and design of machines foundations must be based on standards criteria to ensure the stability. Figure (3) shows the system of forces acting on the foundation [18].

The dynamic response of machine foundations is overseen through the component of soil conditions, vibration characteristics, and foundation shape. According this review category previous studies into three main groups reflecting these governing aspects.

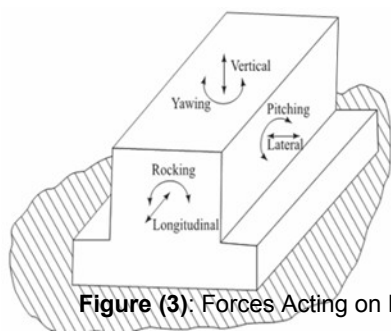


Figure (3): Forces Acting on Machine foundation [18]

4.0 Soil Conditions and Improvement techniques Effects on Machine Foundation Response

Dynamic response of machine foundations is significantly affected by the soil properties on which the foundation rests. Several studies have investigated the dynamic behavior of machine foundations placed on various types of soil: sand, clay, and collapsible soils, etc. The effects of soil type are discussed in this part, with a focus on how the soil responds to vibrations caused by dynamic machines. Modeling of wave propagation is an essential component to dynamic analysis, particularly when dealing with the study of soil foundation interaction. A steel box design to simulate low speed machine operation, using an absorbent layer to reduce wave reflection within the model. The laboratory study consisted of obtaining the soil damping ratio, shear modulus, vibration amplitude, and resonant frequency, as well as performing a dynamic analysis of a circular by measuring the displacement, velocity, and acceleration resulting from harmonic vibrations. The study demonstrated that solid boundaries cause reflection and non-attenuation of waves, while the absorbent layer can reduce this reflection significantly [19].

A numerical analysis performed of machine foundations on four types of cohesionless soils has been examined using PLAXIS 3D finite element software. The study emphasizes the effect of using rigid inclusion piles (non-reinforced concrete columns) for increasing load-bearing capacity, reducing vibration amplitude and achieving a more uniform response under a wide frequency of the forced dynamic load. Comparison of performance between machine foundation rested on rigid inclusion, and also pile and shallow foundations. The analyses revealed that rigid inclusion piles exhibit a more stable response and low dynamic displacement. Additionally, further research could explore the feasibility of enhancing the design of rigid inclusions under machine foundations. The study provides valuable guidance for engineers and designers to optimize the performance of machine foundation systems [20].

Moreover, the performance of machine foundations resting on gypseous soils improved by stone columns, glass fiber, and silica fume, the study highlights how soil improvement techniques influence the transfer of dynamic loading to underground layers of soil focusing on the behavior of the foundation's settlement and pressure distribution, the conditions of the soil were dry and soaked states. A comparative of the experimental results with numerical results by using the 3D Plaxis program has been carried out. Also, the effect of different parameters including amplitude load and frequency was studied [21].

Experimental study was conducted to determine the dynamic characteristics of machine foundations supported by poorly graded soil. Both laboratory and field experiments were conducted to assess the modulus of the subgrade reaction and the response frequency of the base surface. This study analyzed footings resting on ground level and embedded at various depths with and without geosynthetic reinforced. Also, the vibratory response of these foundations was calculated using numerical models that represent linear elastic. Cyclic plate load tests were conducted to estimate the influence of various parameters on the damping capacity of the sand bed. The findings indicated that geogrid reinforcement performed more effectively compared with and without geosynthetic reinforced soil under dynamic excitation [22].

Different geosynthetic materials were explored to improve pressure transmission under dynamic machine foundations, presenting a theoretical model supported by experimental data. There are increases in stress transferred with the increase in the amplitude of load depending on the reinforcement type in the underlying soil layers. The presence of a different geosynthetic reinforcement materials, also the stress level increases when increasing in the amplitude load [23].

An experimental investigation was conducted to study of dynamic loaded for shallow machine foundation on saturated sandy soil made. Model tests were performed with various configurations to evaluate the effectiveness of reinforcement at different depths. Generally, the results validated a major reduction in settlement when geogrid reinforcement was used, mainly in multi-layered soil. Settlement in a medium dense soil was highest at the

beginning of the test, notable also the reinforcement leads to a reduction in settlement, especially in multi-layered soil. The study also examined the relationship between load, displacement, and the possible for liquefaction in sandy soil. The performance of the foundation is influenced by the depth and pattern of the reinforcement layers [24].

The assessment of seismic hazards and the impact of dynamic loads and soil-structure interaction on machine foundation design were reviewed through an analytical review of previous studies related to the seismic analysis and design of machine foundations. This began with a presentation of seismic hazard assessment procedures and the factors influencing them, such as soil properties, machine stiffness, and the impact of seismic loads. Several approaches were identified, most notably the development of more accurate methodologies for seismic hazard assessment using advanced numerical models, in addition to the need for comprehensive experimental studies to validate numerical models and understand the behavior of machine foundations under seismic loads [25].

Machine foundations need expand further study due to dynamic loads which depend on the type of machine and its operating system in addition to the static loads.. A numerical analysis was conducted using ANSYS software to study the dynamic behavior of machine foundations established on improved and unimproved soil reinforced. Natural frequencies and vibration mode were evaluated with the impact of the slope of the surrounding soil. The results indicate that using soil improvement methods are an effective technique for reducing vibration transmission to the surrounding soil. Furthermore, the researcher suggested the using reinforcement materials beneath machine foundations operating within the low-frequency range to reduce vertical and horizontal vibrations [26].

An investigation on how various degrees of saturation in clay soil affect the displacement and stress under machine foundation of circular and rectangular shapes. Experimentally find the displacement and stress variations under different degree of saturation and varied operating frequencies. The study established increasing in the displacement with increasing soil saturation [27].

With the aid of experimental and numerical analyses, the vibrations and soil shear wave velocity were obtained, showing the effects of the water table on the characteristics of the soil and the response of the foundation to dynamic load under different frequencies. The results also showed that the frequency and velocity of the shear wave in the soil changed with variation in water level. There was a good agreement between experimental data and numerical predictions of the measured vibrations that used in foundation design if the exact soil parameters were considered. In addition, the alterations in the water table level can affect frequencies and the damping ratio [28].

An experimental investigation conducted to evaluate the behavior of displacement and stress that developed as a result of force vibrations produced by machine foundation at different frequencies applied on a saturated soft clay.

The impact of embedding depth and load frequency is the focus of the study. Instead, various depth of stresses and pore water pressure is measured under the footing. Results indicated an increase of maximum displacement and increase of pore water pressure with the increase of operating frequencies and that associated with the increase in dynamic load [29].

The dynamic response of a rectangular machine footing established on dry and soaked gypseous soil utilizing experimental techniques. The footing was subjected to amplitudes load generated by a vertical harmonic excitation. The results indicate that the foundation on soaked gypseous soil has smaller displacement response than a dry one, with resonant frequency increased due to soaking. Moreover, larger eccentric mass results in a larger amplitude displacement with a specific frequency and lesser resonant frequency [30].

Furthermore, dynamic analysis was conducted numerically for a rectangular machine footing resting on a saturated clay exposed to four cycles of harmonic loading. The results show that the pore water pressure was developed during the test. The maximum amplitude displacement dissipated rapidly over time due to the presence of the energy-absorbing layer. This study show that the soil type should be taken in consideration when designing machine foundations with varying soil conditions [31].

A numerical model of a prototype scale that used to simulate the experimental setup, proved effective for analyzing machine foundations on saturated soils. The results show a good correlation between the numerical model and experimental study. For a given frequency, the displacement increased with the amplitude of the dynamic force. The small-scale model indicates that the maximum amplitude displacement occurs at the middle of the foundation edge on saturated soil [32].

The dynamic response of soils reinforced with geosynthetic materials investigated through experimental studies using cyclic plate loading. Under dynamic excitation, the study demonstrated that the use of geogrid resulted in more effective performance compared to unreinforced. The behavior of block foundation on surface and those embedded at various depths using geosynthetic materials was analyzed. It was demonstrated that a geogrid at the base enhances the bearing capacity and the recovery of the elasticity. This increases the stability of the system and prevents the foundation failures due to vibrations. There were various dynamic parameters of machine foundations studied through cyclic plate-loading test on geosynthetically reinforced soils. [33]

Previous studies indicate that the soil properties play a significant role in the dynamic response of machine foundations. Variations in soil type, stiffness, damping, and degree of saturation significantly affect the vibration natural and amplitude frequency of the system. Experimental and numerical studies have shown that saturated soils condition, , tend to exhibit increased displacement and lower natural frequencies due to the accumulation of pore water pressure and reduced effective stress. To speech these condition, various soil

improvement techniques have been evaluated, including cement stabilization, geosynthetic reinforcement, solid additives. These methods improve soil stiffness and damping, resulting in a reduction in vibration levels. The reviewed studies show how machine foundations behave under dynamic loads. They all agreed that the increase in saturation results in an increase in displacement under vibration load. From what has been presented, it can be said that there are more studies needed on the dynamic response of foundations subjected to repeated dynamic loading under various water conditions particularly for problematic soils.

The studies reviewed clearly indicate and strong dominance that soil properties play a dominant role in controlling the dynamic response of machine foundations. Vibrations amplitudes and natural frequency are considerably affected by parameters such as soil types, stiffness, damping ratio, and saturation degree of soil. Pore water pressure increases soil displacement and decreases soil stiffness when soils are saturated, while soils improved with the use of geosynthetics or stabilization techniques perform better and at lower vibration levels. Furthermore it is argued that the majority of studies will be performed on sandy soils, because of which there is little research regarding problematic soils (e.g. collapsible or expansive soils). In addition, the combined effect of soil improvement techniques and dynamic loading conditions requires further investigation.

5.0 Dynamic Response of Machine Foundations

Dynamic performance due to the effect of vibration modes and foundation shape on dynamic performance has been widely studied. The shape of machine foundations as well as the vibrations show a critical role in determining dynamic response. The shape of a foundation can change the way it responds to dynamic loads, while the various vibration modes also affect the stability and performance of the structure under the operating conditions.

An experimental study was conducted to investigate the performance of foundation systems for electric generators. Four foundations with support to electrical generators were analyzed. Steady-state horizontal and vertical vibrations tests on all four systems were conducted in the study whereby natural frequencies and amplitude of vibrations were recorded. The experimental values were compared with the required design parameters. The findings showed that the vibration did not damage the housing structures or other adjacent buildings of any of the systems. Settlements and destruction of the generator units were not observed. Also, the segment of critical damping values, which were determined by means of particular analyzing, and it was confirmed that the foundation systems were stable and worked correctly [34].

The analysis of turbo machine foundation, both statically and dynamically, was conducted using ANSYS program. The study simulated a concrete frame foundation supporting a high-speed rotating machine, with specified dimensions for columns, slabs, and raft foundations. The loading and boundary conditions of the structure are fixing the bottom of the raft, and fixing the displacements and

rotations as zero. The dynamic analysis was made in two stages: a frequency analysis to determine natural frequencies and mode shapes. The study also highlights the impact of compression and tension forces on natural frequencies, emphasizing the role of pre-stress. As a result of the modal analysis in case neglecting the mass of the machine minimizes the computed natural frequencies. Adding machine mass gives more realistic frequency values. Machine mass elevation does not influence lateral response [35].

The effect of several parameters on the different vibration response modes of model footings on sand was investigated. A rectangular machine footing with aspect ratios (length/width=1.33) was selected as the model footing to achieve the study's objectives. A model was set to simulate harmonic dynamic loading applied at different operating frequencies with consistent load amplitude. A total of nine cases were conducted on dry sand with relative density 50%. Soil behavior beneath the footing was evaluated by measuring amplitude displacement and strain using a vibration meter. The results indicated that changes in vibration mode had a minimum impact on the final strain values under the same frequency, as well as on the displacement amplitude in the vertical direction. However, the amplitude displacement during rocking vibration was higher than the pitching vibration at the same frequency [36].

In another study the response of circular machine foundation resting on sandy soil to harmonic excitation was examined. A special setup was intended to simulate the vertical vibration of a machine foundation; experimental work was carried out on six footings model placed on dry medium and dense sand were modeled in lab. A footing was represented with a circular steel model subjected to dynamic amplitudes load with different frequencies. The results showed a noticeable increase in the settlement rate for both dry medium and dense sand as the frequency was increasing. Nevertheless, with the same loading and frequencies, a higher relative density of soil corresponds to a decrease in settlement. It was also observed that the displacement amplitude reduced with increase in frequency and relative density. Additionally, only a slight variation in dynamic stress was observed between foundation depths B and 2B. The stress increased as the frequency and the relative density of the sand decreased under the same load amplitude [37].

Similarly, a study for analyzing the block foundation of the machine by considering the coupling take place in between the machine base frame contact with the block foundation at the top and the soil beneath the block foundation at the bottom. The combined effect was evaluated in terms of amplitudes and natural frequency. This produces an elastic systems having a natural period of vibration, and if then unbalanced inertia or other exciting force may occur at a frequency close to the natural frequency of the foundation, causes a resonance lead to excessive vibration. It is evaluated based on a problem of horizontal single cylinder reciprocating; because of the working of the reciprocating machines unbalanced forces are produced so the additional consideration is done other than static analysis is done [38].

To examine the effect of footing shape on dry sand subjected to a machine dynamic loading, a laboratory test was designed to simulated different operating frequencies. In order to evaluate the combination of frequencies of different footing shapes: square, circular, and rectangular footings were tested on medium and dense sand under amplitude load. It was observed that for circular footing, the bearing capacity of the soil was higher than a square and rectangular. Analysis of circular, square and rectangular shapes in three sizes show that the rectangular footing has a lesser dynamic response compared to the circular and square footings [31][36].

Moreover, an experimental technique investigated the dynamic behavior of two close square footings placed on dry and wet collapsible soil. Dynamic loading was imposed on the main footing using a rotating mass-type mechanical oscillator across a range of frequencies, with the objective of evaluating dynamic response, while the other footing was statically loaded. It was found that the amplitude displacement, velocity, and acceleration of the footing under effect of dynamic load for adjacent footings decrease as the distance between these footings increases with greater values in the dry state than soaked state. Additionally, displacement amplitude and velocity increase with operational frequency, whereas acceleration decreases with lower frequency. The dynamic loading effect is more in the dry state [39].

A study examining load cases, displacement, and stress for a LESSAR machine to compare the evaluated frequency ratios and displacements of the foundation. Block and box foundation types for reciprocating machines were explored. Results show that larger foundation areas reduce soil pressure and increase natural frequencies. Vibrations at low frequencies can affect nearby structures. The vertical displacement is greater than the displacements in the other directions for all soil types. Stress distribution varied, with maximum stress in the block foundation and box foundations at the sides, reducing toward the base [40].

Moreover, dynamic analysis of turbo generator machine foundation is essential for confirming requirements for operational reliability and safety. Variables were adopted including foundation's natural frequencies, stiffness, and mass distribution. Vibration characteristics such as amplitude, velocity, and acceleration are analyzed under dynamic loads are produced during the operation of the machine. Additionally, operational parameters include the frequency at which the machine is running, the mass of the rotor, imbalance forces and the shaft elevation. The analysis was done using finite element in SAP2000. The natural frequencies of the foundation were identified by free vibration and harmonic forced vibration analyses to investigate the effect of forces acting on the foundation which were the horizontal dynamic loads. The main results show the necessity of avoid resonance conditions by ensuring that natural frequencies are out the machine's operating range. that modifications in the stiffness and mass of the foundation The main findings are that the resonance conditions should be avoiding by making sure that the natural frequencies are outside the operating range of the machine. Furthermore, the study showed that the stiffness and mass of the foundation can be effectively

control vibration amplitudes within acceptable limits. It came to be concluded also that a well-designed foundation significantly enhances the performance and longevity of turbo-generator machines [41]. Natural frequency and damping ratio of vertically vibrated foundations numerically analyzed and it was shown that various modes of vibrations significantly impact on the dynamic performance of the foundation. The shape of the foundation and vibration mode can alter its dynamic performance [42].

Cone models and experiment data were used to study the dynamic response of machine foundations with a rigid circular shape found on the layered soil subjected to harmonic vertical excitation. The study shown the interaction of foundation geometry in the different layers of soil in order to influence the transmission of vibrations. Field experiments have shown that the intricate stratification of soil and the various forms of foundation create different modes of vibration, which affect the structuring stability. The experiments were conducted on different types of layered soil: one that had the layer of gravel on the top of the in-situ soil, and the other one had the layer of gravel on the top of a rigid base. Seventy two vertical vibration tests with square model footing and Lazan-type mechanical oscillator with the varying key parameters being: top layer thickness, static weight of the foundation and dynamic loads. The resonant frequency and amplitudes generated by the cone model were compare with experiment results showing close alignment [43].

The dynamic response of machine foundations influenced by loading conditions and vibration characteristics. Numerous studies have demonstrated that vibration mode, operating frequency, and amplitude load are controlling parameters that determine the magnitude of the foundation response. Vertical, horizontal, and rocking modes have been shown to produce responses. Resonance conditions have also been stated as an essential consideration, often resulting in excessive displacements and stresses. Future studies need to explore more compound soil conditions, interactions between different shape of foundations at operating conditions. It can be concluded from the above studies that the vibration modes and foundation geometries have a substantial effect on the dynamic performance. The studies review the effects of various factors, such as foundation shape, type of soil, and frequency on the dynamic behavior of the foundation. Vibrations decreased with increasing frequency, reducing settlement and displacement. Also, the interaction between adjoining foundations under dynamic loading was studied.

The vibration characteristics such as frequency, amplitude, and mode of vibration substantiate the dynamic response of machine foundations as emphasized in the reviewed studies. Of these variables, the operating frequency is extremely important as it establishes resonance conditions which can lead to excessive displacement and potential structural instability. It is observed that different vibration modes (vertical, horizontal, and rocking) produce varying responses, with rocking modes often generating higher displacement amplitudes. Furthermore, the interaction between machine mass, foundation stiffness, and soil properties significantly

affects the natural frequency of the system. Several studies highlight the importance of avoiding resonance by ensuring that the natural frequency of the foundation does not coincide with the operating frequency of the machine. Moreover, numerical modeling approaches, have been widely used to predict dynamic behavior; however, these models need validation through experimental studies. However, few studies have investigated the combined effects of vibration mode, soil condition and foundation geometry in actual operating conditions. This indicates a necessity need for more integrated studies to have a better understand the intricate dynamic actions of machine foundations.

6.0 Foundation Shape and Embedment Depth Effects on Machine Foundation Behavior

Settlement analyses of machine foundations under reciprocating loads were conducted using the finite element method to evaluate their behavior in geotechnical engineering. The analysis considered factors such as soil type, foundation shape, foundation location, and the type of dynamic load. Theoretically the study examined the effects of dynamic loads on sandy soil with varying densities and different machine foundations shapes: square, rectangular, strip. The analysis also explored the influence of varying dynamic frequencies on the displacement of the machine foundations. The results indicated that higher values of (L/B) of the foundation lead to a reduction in displacement value for a constant frequency for all the soil conditions. Moreover, the normalized displacement value with increase of the elastic modulus, which shows it tends to the resonance at low frequency. For the square base, this impact is more pronounced than the rectangular and strip foundations [44].

Similarly, interference effects occurred when two adjacent square foundations subjected to a dynamic load were conducted experimentally on closely spaced machine foundations. Various displacement amplitudes were measured by simulating static loads and vertical harmonic loads on one foundation while neglecting it on the adjacent foundation. The model simulations were well validated by parametric wave series, and in most cases, there was a good agreement between predictions and experimental results, supporting the validity of this simplified theory. The results verified the effect of frequency on displacement amplitudes. Dynamic interference effects were observed between closely spaced square machine foundations. Also, the displacement ratio varied with frequency and spacing, with larger displacements occurring when the foundations were closer together [45].

An investigated the dynamic response of a number of machine foundations located on medium and dense sand studied. Nine models were prepared for laboratory tests at different frequencies and shapes of footing circular, square, and rectangular to investigate the effect of footing shape on the dynamic behavior of soils. It was found that the characteristic of footing shape has a pronounced

influence on the bearing capacity of the soil under the dynamic load [46].

Also, the dynamic response of sandy soils in dry conditions with two footing shapes (circular and square) with different relative densities of sand, three embedment depths of foundation and three rates of loading was study experimentally under cyclic dynamic load. Results showed that increasing footing depth reduces settlement when other factors remain constant, and bearing capacity improves with greater foundation depth. Settlement increases over time, peaking at the end of the period load, with partial rebound during the decline phase. The effect of the loading amount depends on sand density, the settlement increases with the loading rate in loose sand and decreases in dense sand [47].

Dynamic response of two adjacent circular and square footings placed on dry and saturated soil analyzed experimentally. The dynamic response of a circular foundation subjected to a dynamic load caused by an adjacent foundation that acts as a vibration source. The adjacent square foundation is stimulated by a known vibration source placed on its surface. The effect of the generated dynamic motion on the nearby foundation, referred to as the second foundation. The results show that increasing the distance between the two foundations leads to a significant decrease in both the vibration amplitude and acceleration of the second foundation. Furthermore, the results indicate that these dynamic responses are more clearer in dry soil than in saturated soil [13].

Dynamic behavior of stepped foundations placed on a medium-density sandy soil surface under the influence of machine vibrations at different operating frequencies was investigated [48]. An experimental study was conducted on four foundation types first a regular foundation essential as a reference and three stepped foundations with different base-to-base ratios (1.0, 1.2, 1.5, and 2.0). The models were subjected to three different operating frequencies of 10, 40, and 70 Hz. The displacement, velocity, acceleration, and stress distribution, were measured at different depths within the soil. The results showed that stepped foundations were more effective than regular foundations in reducing stress and vibration amplitudes, with the stepped foundation at ratio of bottom area to top area of foundation (A_2/A_1) equal to 2.0. Also, the study showed that increasing the area of the lower base improves load distribution and reduces the vibrations. The experimental results compared to the numerical simulations using PLAXIS 3D, the simulation give a good agreement between the numerical and experimental results.

Dynamic analysis of block machine foundations using Barkan's method was carried out by comparing the analytical solution with the results of the finite element method study [11]. The study included three types of reciprocating machines at different operating speeds (150, 250, and 450 rpm) and different soil types: clay (medium, hard, and very hard) and sand (loose, medium, and dense). The effect of soil parameters, particularly the uniform compaction modulus, on the normal frequency and vibration amplitudes of the foundation was examined. The results indicate that increasing the soil stiffness leads

to an increase in the normal frequency and a reduces the vibration amplitudes, while increasing the contact area between the foundation and the soil reduces the dynamic response. The comparison revealed that the finite element method yields lower amplitude values compared to the classical method, indicating that traditional solutions are more conservative.

Recent studies have enhanced the modeling of machines both the static and dynamic properties of machines. Yet, few studies consider the interaction between the machine, supports, foundation, and subsoil. The soil-foundation-support-machine system modeling using finite element method. A sensitivity analysis of the reduction of vibration transmission due to variations in system properties was performed. The results proved that the properties of the soil beneath the foundation and the machine itself are among the most influential factors on vibration transmission. The stiffness and mass of the foundation were also observed to be significant factors affecting the dynamic response. Certainly, the study showed that the impact of supports is relatively less pronounced, although they still play a role in enhancing or weakening the dynamic coupling between system components. The study highlighted the importance of incorporating the interaction between the foundation and the underlying soil into the dynamic analysis, noting that the interrelationship of system properties in real-world applications presents a limitation requiring further investigation [49].

Discussed the effect of the embedment depth of a square foundation which is subjected to dynamic loads imposed by machine foundation's resonant displacement. The authors used experimental measurements of the vertical foundation displacement to validate their calculations. The results showed that the resonant displacement varies significantly with the embedment-depth ratio as vary the dynamic damping and the dynamic stiffness. They also investigated the effects that the size of foundation and the dimensionless frequency on resonant frequency. The results demonstrated a good agreement between the theoretical analysis and the experimental findings regarding the vertical displacement. It was concluded that resonant frequency decreases with increase in embedment depth and foundation dimensions, and decrease in dimensionless frequency. In general, the resonant displacement decreases with increasing embedment ratio and foundation dimensions, and also decreases with decreasing dimensionless frequency, although the effect of embedment is limited at shallow depths.

Conducted field experiments of a square foundation on a clay soil mixed with gravel, tested under vertical vibrations of different loading frequencies. The foundation was embedment at various depths, including partial and full embedment. The results showed that the maximum displacement and resonant amplitude decreased with increasing embedment depth of foundation. The dynamic stiffness and damping ratio increased with increasing embedment depth, while the resonant frequency changes slightly. These findings confirm that the performance of machine foundation can

be significantly improved by increasing the foundation embedment depth [51].

Foundation type as well as embedment depth, has a significant effect on the dynamic behavior of machine foundations. Previous research has indicated that foundation mass, shape and stiffness directly influence on the natural frequency and damping of the system. The embedded foundations show better improved vibration performance compared to surface foundations. Further embedment increases damping and reduce vibration amplitudes, particularly in problematic soils. The effects of embedment depth and foundation type depend in an important way on soil and loading conditions.

The reviewed studies demonstrate that foundation geometry and embedment depth strongly influence on the dynamic behavior of machine foundations. Foundation shapes, size, and depth parameters immediately influence the stiffness, damping attributes, and have similar natural frequencies of the machine. In general, it has been noticed that the circular and larger footprints have better load distribution with reduced vibration response compared to other shapes. Additionally, increasing embedment depth improves the overall stability of the foundation by enhancing soil confinement, increasing damping, and reducing displacement amplitudes. Several studies confirm that deeper foundations exhibit lower vibration amplitudes and improved dynamic performance, particularly in soft or saturated soils. However, the effectiveness of embedment depth depends on soil conditions and loading characteristics. Despite these findings, limited research has been conducted on the combined influence of foundation geometry, embedment depth, and complex soil conditions under varying dynamic loads. This highlights the need for further studies focusing on integrated analysis of these parameters to optimize machine foundation design.

7.0 Summary of Findings and Future Research Directions

The dynamic performance of machine foundations is essentially overseen by the coupled interaction between soil properties, foundation geometry, and loading conditions. This review demonstrates that soil parameters mainly stiffness, damping ratio, and degree of saturation are the primary determinants of vibration response, directly influencing both amplitude and natural frequency. Soft and saturated soils, due to excess pore water pressure and reduced effective stress, exhibit significantly higher displacements and lower stiffness, thereby necessitating appropriate ground improvement strategies.

Treatment techniques such as rigid inclusions and geosynthetic reinforcement have shown considerable effectiveness in enhancing soil stiffness and attenuating vibration-induced settlements. In parallel, foundation geometry plays a decisive role; Choosing the most suitable foundation shape generally provide more favorable dynamic performance, while increased embedment depth contributes to improved confinement, higher stiffness, and reduced vibration amplitudes. From a design perspective, the avoidance of resonance remains a

critical criterion, given the high sensitivity of system response to the proximity between operating and natural frequencies. Moreover, the study of effects between adjacent foundations, Even with substantial progress, the recent of research remains limited in giving long term cyclic loading effects, groundwater conditions, and heterogeneous soil profiles.

The studies reviewed can be classified into three primary categories: soil improvement, dynamic behaviour of foundations, and soil–structure interaction. Studies of soil improvement methods (geogrid reinforcement, rigid inclusions, stone columns and absorbent layer) reduced displacement, settlement, wave reflection and vibration transmission. These studies, although numerous, mostly studied each improvement method separately with little direct comparison of different techniques under the same loading and soil conditions. most of these studies considered each improvement method in isolation and including limited direct comparisons between the different techniques under identical loading and soil conditions.

Research on the dynamic response of foundations shows that foundation reaction is a function of operating frequency, resonant frequency, foundation configuration, soil density and effect of embedment depth. Various studies have indicated that the displacement and settlement increases with increase in excitation frequency. Still, many of these studies dealt exclusively with isolated footing behavior and did not represent the complete machine–foundation–soil systems that indicate machine foundations should not be treated as independent structural components. Natural frequency and dynamic response is greatly affected by the interaction between soil stiffness, foundation mass, damping, vibration modes and machine characteristics.

More experimental and numerical research should be developed for better soil–foundation–machine systems in the future. These studies must investigate for displacement, settlement, resonance, damping, stress distribution and frequency response under realistic operational conditions. These studies need to assess displacement, settlement, resonance, damping, stress distribution and frequency response during realistic operating scenarios. Tables (A, B, C) summarize what is explained in the three paragraphs above.

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Table A: Soil Improvement Studies

Ref	Authors	Year	Technique	Soil Type	Method	Key Parameters	Main Findings
[19]	Abood et al.	2024	Absorbent layer	Granular soil	Experimental	Damping ratio, shear modulus, resonance	Reduces wave reflection; improves damping
[20]	Khalil	2023	Rigid inclusion piles	Cohesionless soils	Numerical (PLAXIS 3D)	Displacement, frequency response	Reduces displacement and improves stability
[21]	Al-Azzawi	2023	Stone columns + additives	Gypseous soil	Exp. + Num.	Settlement, pressure, frequency	Enhances load transfer and reduces settlement
[22]	Parmar et al.	2021	Geogrid reinforcement	Sand	Exp. + Num.	Subgrade modulus, damping	Improves dynamic performance
[23]	Karim et al.	2021	Geosynthetics	Reinforced soils	Theoretical + Exp.	Stress distribution, load amplitude	Improves stress behavior under loading
[24]	Karim et al.	2020	Multi-layer geogrid	Saturated sand	Experimental	Settlement, liquefaction	Reduces settlement significantly
[26]	Ahamed et al.	2022	Soil reinforcement	Reinforced soil	Numerical (ANSYS)	Natural frequency, modes	Reduces vibration transmission
[31]	Al-Wakel et al.	2011	Energy-absorbing	Saturated clay	Numerical	Pore pressure, displacement	Dissipates vibration energy

			layer					
[33]	Parmar et al.	2021	Geogrid	Reinforced soils	Experimental	Bearing capacity, stiffness	Improves stability and stiffness	

Table B. Dynamic Behavior of Foundations

Ref	Authors	Year	Foundation Type	Soil Type	Method	Key Parameters	Main Findings
[30]	Al-Busoda & Alahmar	2014	—	Gypseous soil	Experimental	Resonance, amplitude	Reduces displacement, increases resonance
[32]	Al-Wakel et al.	2014	—	Saturated soil	Exp. + Num.	Displacement, frequency	Displacement increases with load amplitude
[34]	Vlad & Vlad	2004	Generator foundation	—	Experimental	Frequency, amplitude, damping	System stable with adequate damping
[36]	Abdul Kaream et al.	2020	Rectangular footing	Sand	Experimental	Mode, displacement, strain	Rocking > pitching effect
[37]	Abdul Kaream et al.	2020	Circular footing	Sand	Experimental	Frequency, density	Settlement increases with frequency
[38]	Bhashakhetre & Marve	2020	Block foundation	—	Analytical	Natural frequency, resonance	Resonance condition identified
[39]	Ibrahim & Zakaria	2019	Adjacent footings	Collapsible soil	Experimental	Spacing, frequency	Response decreases with spacing
[40]	Patel et al.	2017	Block/box foundations	Various	Analytical/ Numerical	Stress, displacement	Larger area reduces pressure
[43]	Pradhan et al.	2008	Circular square /	Layered soil	Exp. + Model	Layer thickness, resonance	Soil layering strongly affects response
[44]	Alhasso et al.	—	Various footings	Sand	Numerical (FEM)	L/B ratio, frequency	Increasing L/B reduces displacement
[45]	Swain & Ghosh	2015	Adjacent foundations	—	Experimental	Spacing, frequency	Dynamic interference increases with proximity
[46]	Abdul Kaream et al.	2020	Various shapes	Sand	Experimental		Shape significantly affects behavior
[47]	Fattah et al.	2017	Circular/square	Sand	Experimental	Embedment depth	Depth reduces settlement
[48]	Al-Obaydi et al.	2025	Stepped foundation	Sand	Exp. + Num.	Shape, frequency	Reduces vibration and stress
[50]	Najm et al.	2022	Square foundation	—	Exp. + Analytical	Embedment ratio	Reduces displacement and resonance
[51]	Mbawala et al.	2017	Square foundation	Clay + gravel	Field Exp.	Depth, damping	Increases stiffness and damping

Table C. Soil–Structure Interaction (SSI)

Ref	Authors	Year	System	Method	Key Parameters	Main Findings
[25]	Noman & Albusoda	2023	General SSI	Review	Seismic response	Need advanced validated models
[28]	Cremaschi et al.	2021	Soil system	Exp. + Num.	Wave velocity, water level	Water level affects damping
[29]	Abdulrasool et al.	2021	Saturated system clay	Experimental	Pore pressure, frequency	Higher frequency increases response
[35]	Ahmed et al.	2023	Machine + raft	Numerical (ANSYS)	Mass, modes	Machine mass is critical
[41]	Jayarajan & Kouzer	2014	Turbo foundation	Numerical	Stiffness, resonance	Avoid frequency overlap
[42]	Ahn et al.	2011	Vibrating foundation	Numerical	Modes, damping	Modes strongly influence response
[49]	Dunaj & Archenti	2024	Soil–machine system	Numerical (FEM)	Interaction, stiffness	Soil + machine dominate behavior

